

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network
Security **COURSE CODE:** CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I M.S. Divyadharsan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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Course Code: CSA5116

Course Name: Cryptography

Date: 6/10/26

CO1 - AT1

Application based questions.

3. Scenario: Secure messaging system. A company deploys a Keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
- a) Demonstrate how the encryption process transforms the message step by step.
- b) Explain how the repeating keyword influences each character of the ciphertext.

Ans: a) Encryption process using Keyword Cipher

Message : H E L L O

Keyword : K E Y

Using a Vigenere-style Keyword encryption:

Message H E L L O

Keyword K E Y K E

Ciphertext R I J V S

So, HELLO $\rightarrow$ RIJVS

Step by step

$$\rightarrow H + K = R$$

$$\rightarrow E + E = I$$

$$\rightarrow L + Y = J$$

$$\rightarrow L + K = V$$

$$\rightarrow O + E = S$$

b) Effect of repeating Keyword

$\rightarrow$ The Keyword KEY is repeated to match the message length.

KEY KEY

$\rightarrow$ Each message character is the message encrypted using the corresponding keyword character. Therefore, the repeating keyword determines how each character of the ciphertext is produced.

4. Scenario: Suspicious Digital Asset Investigation
During a digital forensic audit, investigators suspect that a company logo file may

Contain hidden Confidential information embedded within it. The file appears normal upon casual inspection.

- a) Analyse How investigators could detect the presence of hidden data with the image.
- b) Recommend two techniques that could strengthen the security of such hidden communication against detection or extraction

Ans: a) Detecting hidden data in an image

→ Investigators can:

- 1) Check the image's metadata for unusual or suspicious information.
- 2) Use steganography detection tools to analyse pixel patterns and identify hidden data.
3. Compare the file size and structure with a normal copy of the same image
4. Examine the file for unusual appended data or abnormal file structure.

b) Two techniques to strengthen hidden communication.

- 1) Encryption before embedding - Encrypt the Confidential message before hiding it in the image.
 - 2) Steganography with randomized embedding - Distribute hidden data across pixels using a randomized pattern, making detection more difficult.
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